Civics Group | Reg Number

Candidate's Name ______________________________

MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations
Higher 2

H2 BIOLOGY 9648/01

Paper 1 Multiple Choice

27 September 2013
1 hour 15 minutes

Additional Materials: OMR

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

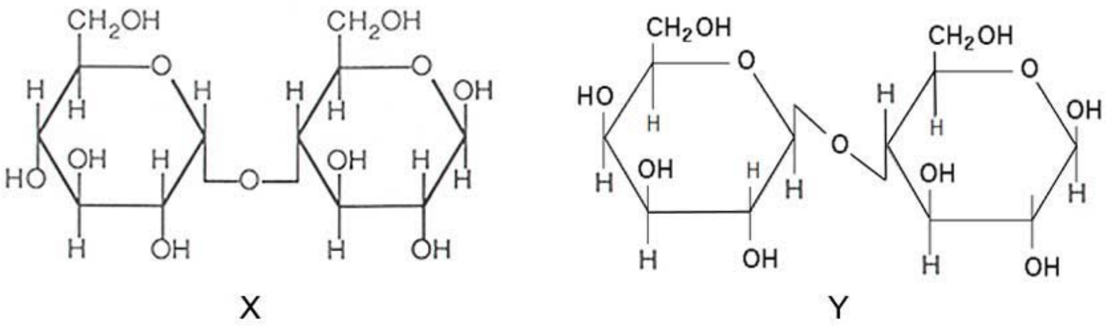
Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, civics group and register number in the spaces at the top of this page and on the OMR.

There are **40** questions in this section. Answer **all** questions. For each question, there are four possible answers labelled **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate OMR answer sheet.

This paper consist of **25** printed pages

[Turn over]

1 The figure below shows two disaccharides, **X** and **Y**.



Which statements correctly describe **X** and **Y** as shown in the figure?

1 Glucose monomer(s) that make up **X** contains both α-glucose and β-glucose, while the glucose monomer(s) that make up **Y** contains only β-glucose.

2 The glycosidic bond that is formed in **X** is α-1,2 glycosidic bond.

3 The glycosidic bond that is formed in **Y** is similar to the glycosidic bond that is found in cellulose.

4 The general formula of a monomer is $(CH_2O_2)n$.

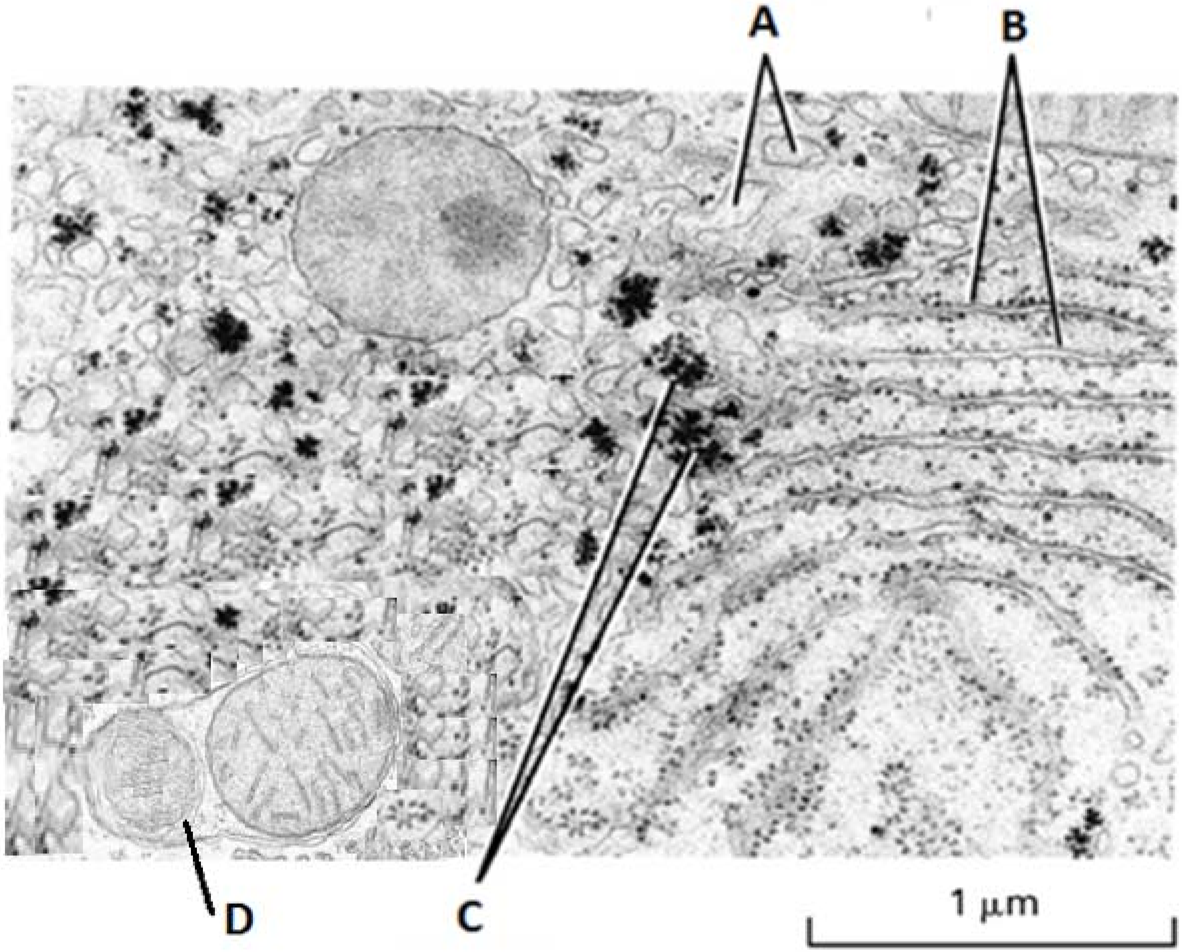
A 1 and 3 only

B 2 and 4 only

C 1, 3 and 4 only

D 1, 2, 3 and 4

2 The electron micrograph below shows a liver cell.



Which statements correctly describe the labelled structures?

1 Structure **A** transports proteins from Structure **B** to Golgi Apparatus.

2 Proteins enter the lumen of Structure **B**, where they undergo chemical modifications such as glycosylation.

3 When Benedict's reagent is added to Structure **C**, the Benedict's reagent would form a brick red precipitate.

4 The process shown in structure **D** is autolysis.

A 2 only

B 1 and 2 only

C 2 and 3 only

D 2, 3 and 4 only

3 Which one of the following statements about the cell surface membrane is true?

A It is totally impermeable to water due to the large number of hydrophobic molecules it contains.

B It is a bilayer, 7 – 10nm thick, containing two layers of intrinsic proteins sandwiched between two layers of phospholipids.

C It sometimes contains cholesterol which regulates membrane fluidity by preventing it from becoming too fluid at high temperatures or too rigid at low temperatures.

D Glycolipids and glycoproteins are biological markers which act as antibodies to destroy foreign antigens of the infecting pathogens.

4 The table below summarises the results from experiments on the rate of absorption of certain monosaccharides by pieces of mammalian intestine.

Cyanide inhibits cytochrome c oxidases.

		Relative rate of absorption by intestine (arbitrary units)	
		In physiological saline	In physiological saline with added cyanide
Hexoses	Glucose	1.00	0.33
	Galactose	1.10	0.53
	Fructose	0.43	0.37
Pentoses	Xylose	0.30	0.31
	Arabinose	0.29	0.29

Which of the following statements are consistent with these results?

1 Active transport is not involved in the uptake of pentoses.

2 Xylose and arabinose enter by osmosis.

3 Aerobic respiration is necessary for the maximal rate of uptake of glucose.

4 Smaller molecular weight sugars are absorbed more easily.

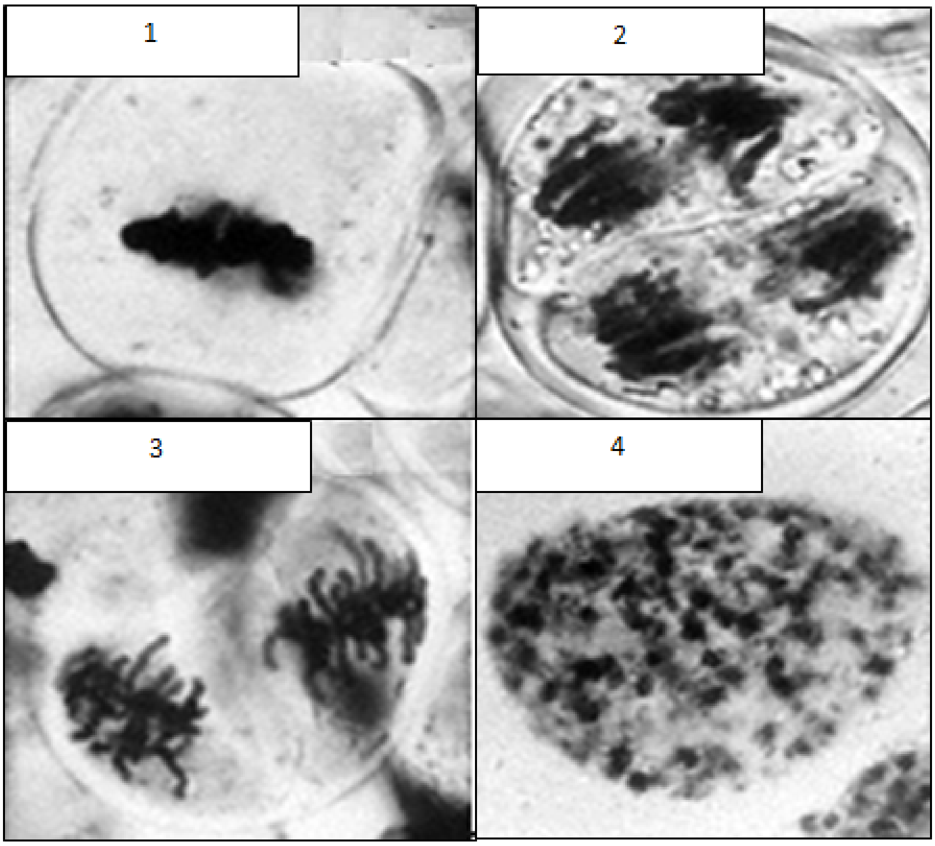
A 1 and 3 only

B 1, 2 and 3 only

C 1 and 4 only

D 2 and 4 only

5 The following figure shows nuclear division occurring in a lily plant.



Which of the following shows the correct sequence of events?

A 1 > 4 > 3 > 2

B 1 > 3 > 4 > 2

C 4 > 1 > 3 > 2

D 4 > 2 > 3 > 1

6 A biologist counted 2,500 cells from an embryo on a microscope slide and recorded the following data.

Stage	**Number of cells**
Prophase	125
Metaphase	50
Anaphase	50
Telophase	25
Interphase	2,250

From the table above, it could be reasonably concluded that

A the cells are undergoing differentiation.

B the cells are cancerous.

C the cells are dividing randomly.

D interphase is the longest stage of nuclear division.

7 Which statement about the cell cycle is **not** correct?

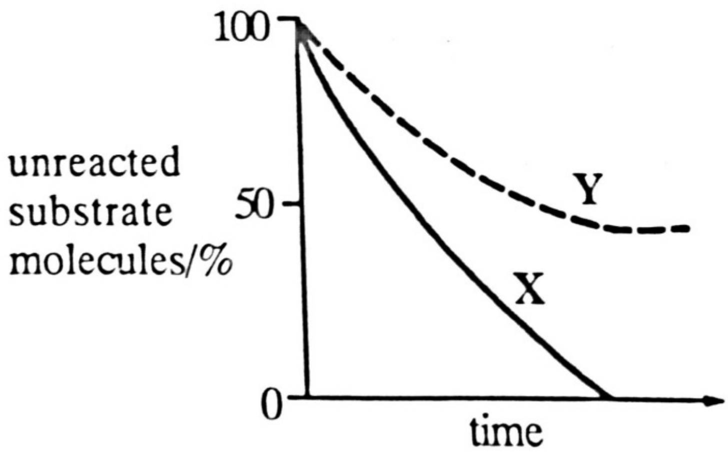
A The G_2 checkpoint checks for DNA damage or mutation.

B Rapidly dividing cells have a shorter G_1 phase than non-dividing cells.

C Protein synthesis mainly occurs during the G_1 and G_2 phases of the cell cycle.

D Dysregulation in cell cycle checkpoints causes chromosomal translocations and cancer.

8 Curve **X** represents the course of an enzyme-catalysed reaction under optimum conditions. Curve **Y** shows the action of the same enzyme on the same substrate but with one alteration to the reaction conditions.



Which of the following factors, operating to a constant extent throughout the experiment, could give the results shown by curve **Y**?

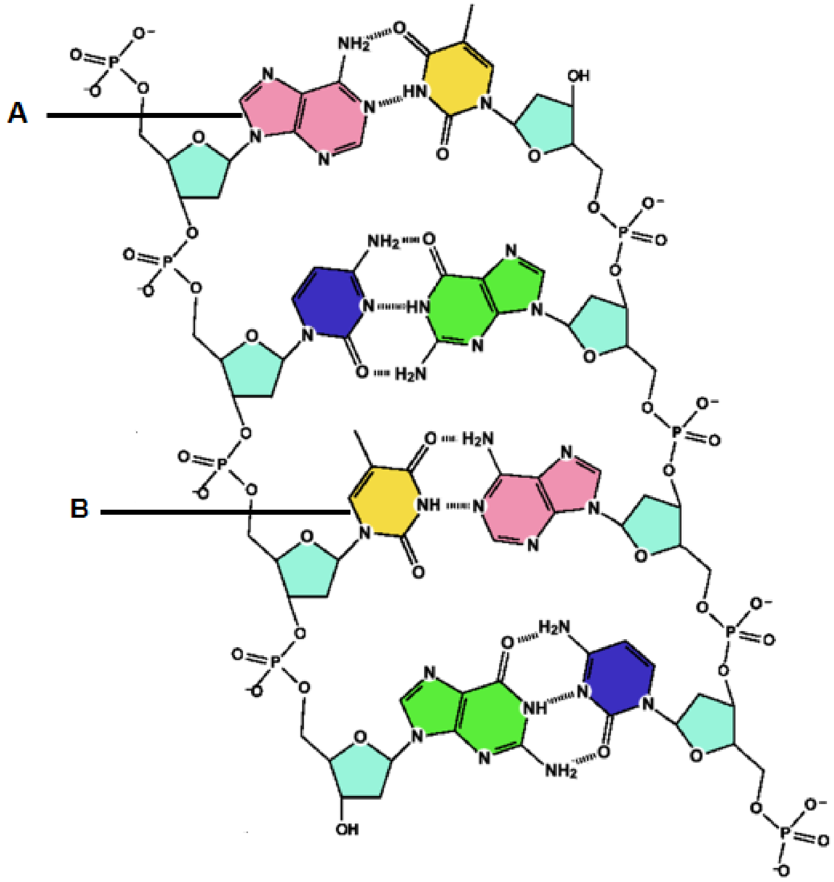
A Addition of a competitive inhibitor

B Addition of an end-product inhibitor

C An increased substrate concentration

D A lower temperature

9 The figure below shows a DNA molecule.



Which statements correctly describe the polynucleotide?

1 The structure labelled **A** corresponds to a purine, while the structure labelled **B** corresponds to a pyrimidine.

2 The antiparallel nature of DNA double helix aligns the DNA strands so that phosphodiester bonds can form properly between the nitrogenous bases from opposite strands.

3 Distance between adjacent deoxyribonucleotides is 3.4 Å and with 10 deoxyribonucleotides per turn. (Note: 10 Å = 1 nm)

4 The wound DNA double helix consists of alternating major groves and minor groves along its axis, that is essential for the binding with proteins.

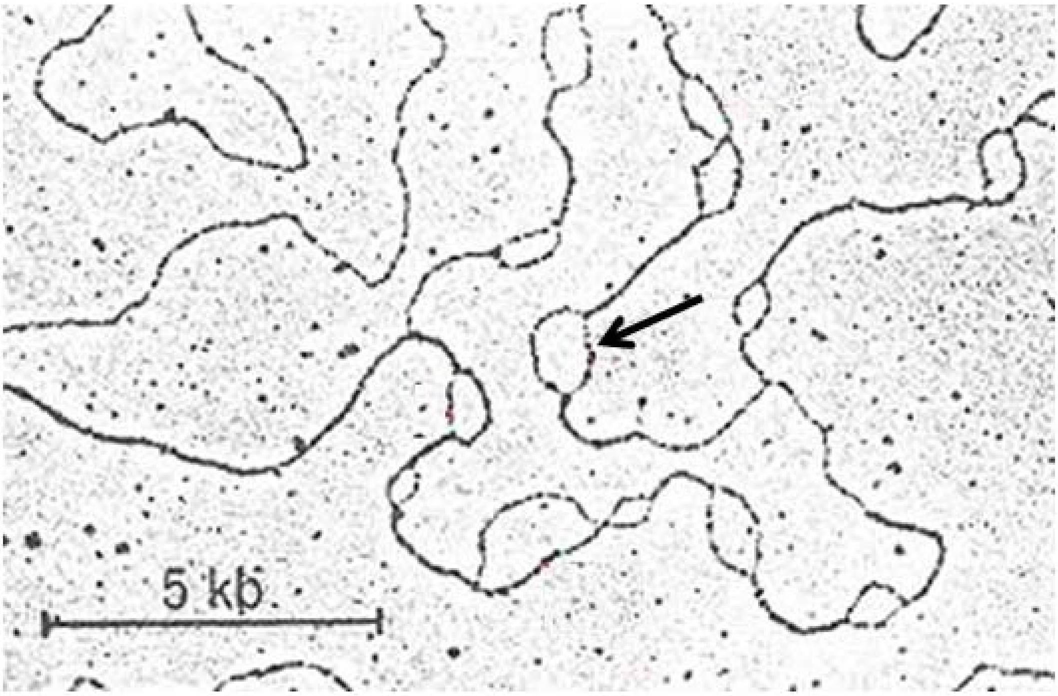
A 1 only

B 1 and 2 only

C 3 and 4 only

D 1, 3 and 4 only

10 The figure below shows structures that are formed along a DNA molecule of a eukaryotic cell during S phase of the cell cycle.



Which statement(s) correctly describe the labelled structure and the process that is taking place?

1 Many such structures can also be found in prokaryotic DNA.

2 DNA replication requires primase to synthesize primers to provide free 3' OH for the elongation of daughter strands by RNA polymerase.

3 Eukaryotic chromosomes face the end-replication problem as they are linear. Prokaryotic chromosomes do not have the end-replication problem as they are circular.

4 The end-replication problem occurs only on the lagging strands.

A 1 and 2 only

B 1 and 3 only

C 3 only

D 3 and 4 only

11 The electron micrograph below shows several labelled structures present in a mitochondrion.

Which of the statements below correctly describes the labelled structures?

1 The structure labelled **A** is the polypeptide chain.

2 The structures labelled **B** are polyribosomes. Each ribosome consists of a 50S large subunit and a 30S small subunit.

3 The structure labelled **C** is the 3' end of template DNA strand.

4 The structure labelled **C** is the 5' end of the mRNA strand.

A 1 and 2 only

B 1 and 4 only

C 2 and 3 only

D 1, 2 and 4 only

12 Which of the following statements correctly describes the type of mutation that gives rise to sickle cell anaemia?

A Deletion of a codon resulting in the deletion of amino acid phenylalanine at position 508 of the polypeptide chain.

B Deletion of a base pair resulting in the deletion of amino acid phenylalanine at position 508 of the polypeptide chain.

C Base pair substitution of thymine by adenine on the non-template strand of DNA causes a change in amino acid from glutamic acid to valine.

D Base pair substitution of thymine by adenine on the template strand of DNA causes a change in amino acid from glutamic acid to valine.

13 Which of the following statement correctly describes introns and/or exons in a eukaryotic cell?

A Mutations to introns have absolutely no effect on the primary structure of protein.

B Different combinations of exons and introns allow more than one type of proteins to be coded for by a gene.

C Exons are always translated into proteins.

D Single base-pair deletion on the first exon of a gene is likely to be more harmful than one which occurs on the last exon of a gene.

14 The following statements describe a ribosome and a telomerase.

1 The genes coding for the ribosomal proteins are found in the nucleolus whereas the genes coding for the spliceosomal proteins are found in other parts of the nucleus.

2 A ribosome is involved in condensation reactions whereas a telomerase is involved in hydrolysis reactions.

3 Both function in the cytosol.

4 In performing their functions, both involve the formation of hydrogen bonds.

5 Both are found in the cells which constitute the inner cell mass of a blastocyst.

6 Both are ribonucleoproteins.

Which of the statements are true?

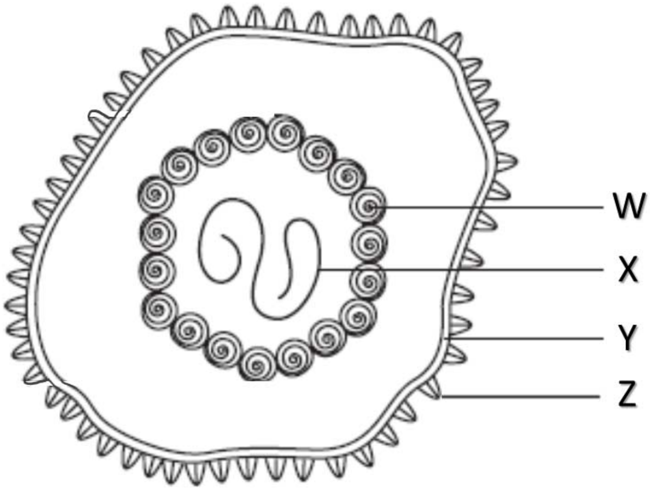
A 1, 2 and 3

B 1, 4 and 5

C 2, 3 and 6

D 4, 5 and 6

15 The figure below shows the structure of a virus.



Which of the following matches the functions of structures **W** – **Z**?

	W	X	Y	Z
A	Ensures the integrity of the viral genome is maintained	Entry of virus into host cell	Specificity of host cell	Assembly of viruses
B	Ensures the integrity of the viral genome is maintained	Assembly of viruses	Entry of virus into host cell	Specificity of host cell
C	Specificity of host cell	Assembly of viruses	Ensures the integrity of the viral genome is maintained.	Entry of virus into host cell
D	Assembly of viruses	Ensures the integrity of the viral genome is maintained.	Entry of virus into host cell	Specificity of host cell

16 To date, more than 10 different strains of influenza virus (e.g. H1N1, H2N3, H5N1, H7N9 and so on) have been documented.

Which of the following structural characteristic of influenza virus makes this possible?

A Single-stranded RNA as its genetic material

B Presence of an envelope that is derived from the host cell

C Eight separate segments of genetic material

D Presence of error-prone reverse transcriptase within the virus

17 During heat-shock transformation, competent *E. coli* cells take up plasmid DNA which contains ampicillin-resistant gene. Successfully transformed cells are allowed to divide for several generations in a medium without ampicillin. It is observed that some of the *E. coli* cells are no longer resistant to ampicillin.

Which of the following best accounts for the observation?

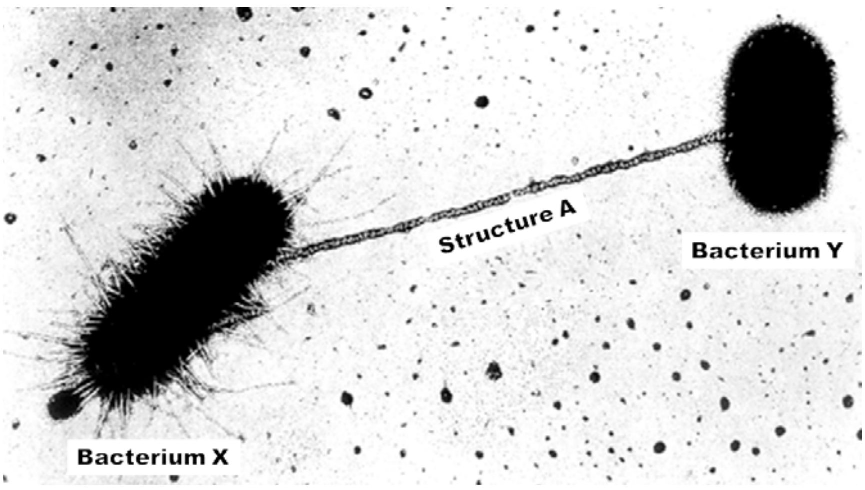
A Error during DNA replication causes a loss-of-function mutation to the ampicillin-resistant gene.

B Incorporation of the plasmid DNA into the heterochromatin region of the bacterial genome renders the ampicillin-resistant gene inactive.

C Restriction endonucleases present in *E. coli* cells recognize specific sequences on the plasmid DNA and hydrolyze it into fragments.

D Unequal distribution of plasmid during binary fission results in some daughter cells without plasmids.

18 The following shows a process taking place among prokaryotic cells.



Which of the following best describes the process?

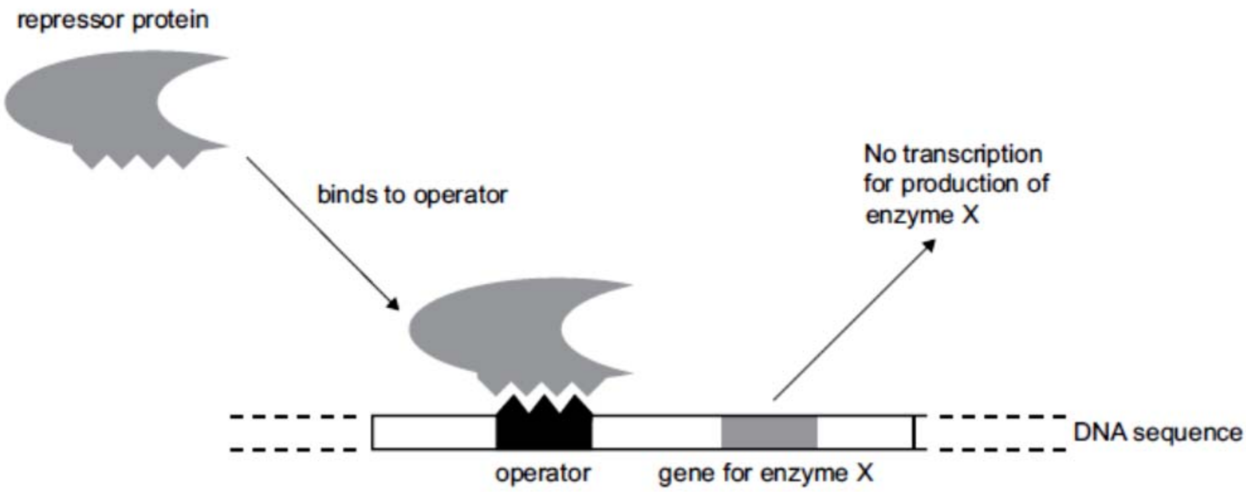
A A polynucleotide can be found within Structure **A**.

B Structure **A** is made up of protein subunits that are coded for by the genes carried on the chromosomal DNA in Bacterium **X**.

C At the end of the process, Bacterium **Y** will be genetically identical to Bacterium **X**.

D DNA replication is taking place only in Bacterium **X** but not in Bacterium **Y**.

19 In a certain species of bacteria, the gene coding for enzyme **X** is not transcribed when the repressor protein binds to the operator gene.



Which protein is prevented from functioning during this binding?

A Enzyme **X**

B TATA-binding protein

C RNA polymerase

D Repressor protein

20 A fisherman was surprised to catch a fish which had no scales (nude). To investigate the origin of this phenotype the nude fish was mated several times to fish with scales and the result of each cross was recorded. In the crosses of nude with scaled, a third phenotype appeared, which was later called linear. The linear phenotype has only a single line of scales down one side of the body.

The outcomes of these crosses are shown in the table.

Cross	Parents	Offspring phenotype and ratio
1	scaled x nude	all offspring linear
2	linear x linear	1 scaled : 2 linear : 1 nude

Which of the following is conclusive from the above data?

A There is incomplete dominance between the nude and scaled phenotype.

B The environment is the reason for the loss of scales in the nude fish.

C All of the linear fish are homozygous.

D The nude fish are heterozygous.

21 Three gene loci in mice are shown below.

Locus 1 Coat colour	**Locus 2** Tail appearance	**Locus 3** Coat appearance
agouti albino	kinky straight	non-frizzy frizzy

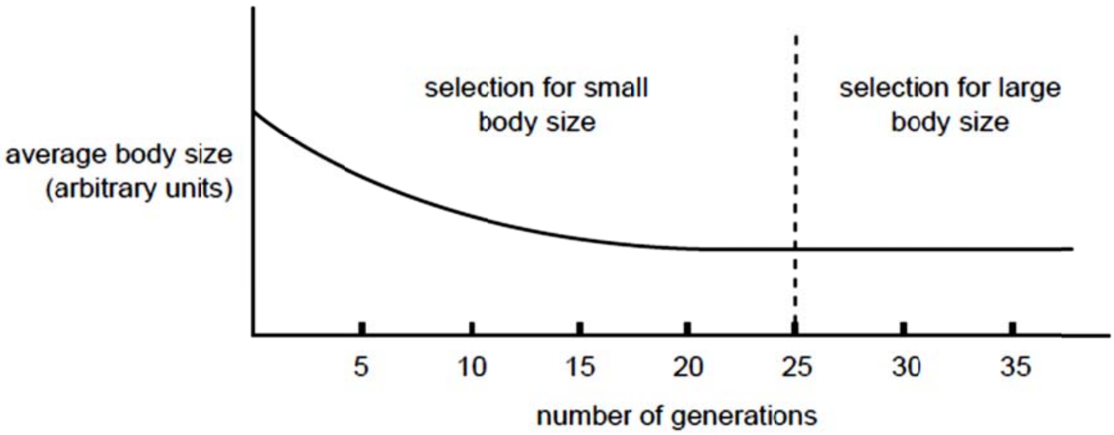
Crosses involving two loci at a time were set up and their outcomes are shown in the table below.

Parents (pure breeding)	**F1**	**Offspring of a test cross of the F1**	
cross 1 agouti, non-frizzy coat x albino, frizzy coat	agouti, non-frizzy coat	agouti, non-frizzy coat albino, frizzy coat agouti, frizzy coat albino, non-frizzy coat	44 46 5 5
cross 2 agouti, straight tail x albino, kinky tail	agouti, kinky tail	agouti, straight tail albino, kinky tail agouti, kinky tail albino, straight tail	23 27 24 26

Which of the following statement is true about **cross 1** and **cross 2**?

	Cross 1	**Cross 2**
A	The frequency of crossing over between the locus 1 and locus 3 is 10%	The agouti coat and kinky-tailed offspring of the test cross of the F1 are heterozygous at both loci.
B	Locus 1 and locus 3 undergo independent assortment.	The albino coat and straight-tailed offspring of the test cross are pure breeding.
C	Locus 1 and locus 3 are located on the same chromosome.	The F1 mice were test crossed with agouti coat and straight-tailed mice.
D	The interaction between locus 1 and locus 3 is an example of epistasis.	Locus 1 and locus 2 undergo independent assortment.

22 The graph below shows the results of a breeding experiment with the vinegar fly *Drosophila melanogaster*. In each of the first 25 generations the smallest flies were selected to produce the next generation. After 25 generations the selection was reversed. From generation 25–35 the largest flies were chosen to breed the next generation.



Which of the following can be concluded from the result of the experiment?

A The flies from generation 25 onwards are phenotypically identical.

B Body size is inherited in a polygenic manner.

C Body size is controlled by one gene with more than two alleles.

D Body size is a sex-linked trait.

23 In *Primula*, the dominant allele, **A**, of a gene for anthocyanin pigment production gives magenta-coloured flowers. The recessive allele, **a**, gives yellow-coloured flowers. The dominant allele, **B**, of another unlinked gene inhibits the effect of allele **A**, while the recessive allele, **b**, has no effect. Plants in which allele **A** is inhibited by allele **B** have yellow flowers.

A yellow-flowered plant is known to be homozygous recessive at the **A/a** locus. The genotype at the **B/b** locus is unknown.

Which of the following genotype of the test cross will reveal the genotype of the yellow-flowered plant?

A aabb **B** AABB **C** AAbb **D** aaBB

24 One of the human blood groups is the MN group. There are two alleles, $\mathbf{L^M}$ and $\mathbf{L^N}$, at this gene locus which determine the presence of an antigen, M or N, on the surface of the red blood cells. The heterozygote $\mathbf{L^ML^N}$ has a different phenotype from each of the homozygotes.

The frequencies of the phenotypes of the **MN** blood group were measured in a European population. Of the 100 individuals sampled, 40 were blood type **M**, 20 were blood type **MN** and 40 were blood type **N**.

Which of the following is conclusive from this data?

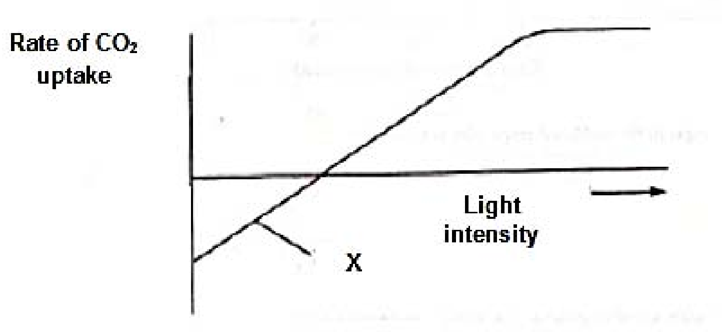
A The frequency of the $\mathbf{L^N}$ allele in the European population is 0.3.

B The frequency of the $\mathbf{L^M}$ allele in the European population is 0.5.

C There are 60 $\mathbf{L^M}$ alleles in the European population.

D There is a total pool of 200 alleles at this locus in the European population.

25 In the graph below, the rate of CO_2 uptake by plant cells is shown to vary with increasing light intensity.



Which of the following is true at point **X**?

A The plant is photosynthesising.

B Respiration equals photosynthesis.

C CO_2 is a limiting factor.

D There is not enough light for photosynthesis to have commenced.

26 Removal of the source of carbon dioxide from photosynthesising chloroplasts results in rapid changes in the concentration of certain chemicals. Which one of the following represents the correct combination of concentration changes?

	ATP	***ribulose bisphosphate***	***glycerate-3-phosphate***
A	decreases	decreases	increases
B	decreases	increases	no change
C	increases	increases	decreases
D	increases	no change	increases

27 Both glucose and appropriate enzymes are necessary for the process of glycolysis to begin. Which additional compound must also be present?

A acetylcoenzyme A

B ATP

C pyruvate

D reduced NAD

28 What happens to most of the reduced NADP molecules in cell metabolism?

A They act as oxidising agents in glycolysis.

B They are oxidised in thylakoid membrane for ATP formation.

C They are oxidised in the Calvin cycle.

D They combine with succinic acid as part of Krebs cycle.

29 A population's gene pool was found to have remained unaltered for many generations. Which of the following conditions must have existed in the population?

A mating had always been random

B genetic drift had often occurred

C inbreeding had been commonly practised

D certain alleles are at selective advantage

30 Competition, isolation and selection are thought to be events that take place during the formation of new species. In which order did these events take place in the process of speciation to form 13 species of Darwin's finches?

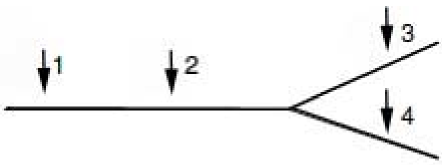
A competition, selection, isolation

B selection, isolation, competition

C isolation, competition, selection

D competition, isolation, selection

31 The dendrite end of the axon is stimulated at the site '1' (see figure below). The excitation is transferred from site '1' to '2' and then to '3' and '4'. The excitation is measured at these sites. Which statement of impulse frequencies (I) measured at these sites is correct?



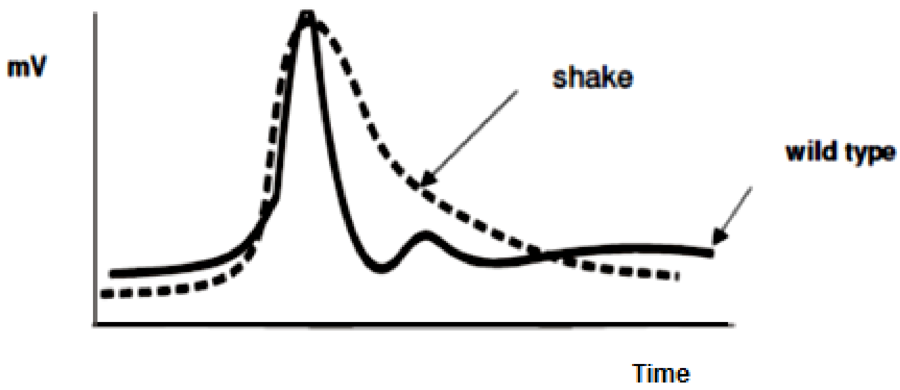
A I(1) > I(2) > I(3), I(3) = I(4), I(3) + I(4) = I(2).

B I(1) < I(2) < I(3), I(3) = I(4).

C I(1) = I(2) > I(3), I(3) = I(4), I(3) + I(4) = I(2).

D I(1) = I(2) = I(3) = I(4).

32 *Drosophila* flies homozygous for the *shake* mutation are extremely sensitive to diethyl ether that causes convulsions in homozygous individuals. Convulsions are caused by abnormalities in nerve impulse conduction as shown in the graph below. Which of the following structures is impaired in the *shake* mutations?



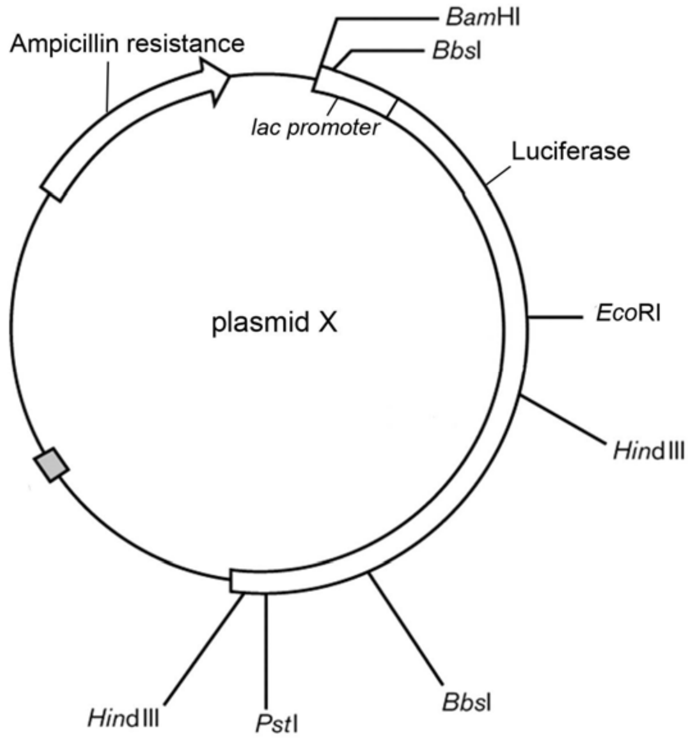
A Na^+ channels

B K^+ channels

C Ca^{2+} channels

D Na^+/ K^+ pump

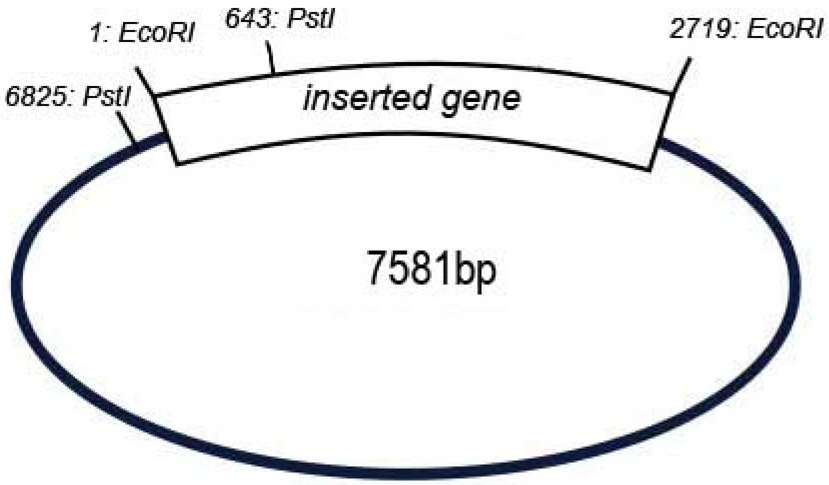
33 Plasmid **X** can serve as a vector for the insertion of genes to be cloned.



Which of the following options will allow the selection of the colonies containing the recombinant form of plasmid **X**?

	Selection medium	Phenotype of colonies that contain the inserted gene
A	Containing ampicillin and lactose	White colonies
B	Containing ampicillin and luciferase	Colonies that emit light
C	Containing ampicillin, lactose and luciferin	White colonies
D	Containing ampicillin, lactose and luciferin	Colonies that emit light

34 A scientist planned to insert a gene of approximately 2719 base pairs into a plasmid vector using restriction enzyme *Eco*RI. The diagram below shows the recombinant plasmid with the gene in the desired orientation.



Restriction mapping can be used to check the orientation of the gene inserted into a plasmid. The recombinant plasmid is cleaved with restriction enzymes(s) and the fragments are run on an electrophoresis gel to ascertain their size.

Which of the following results will confirm that the gene was not inserted in the desired orientation?

	Restriction enzyme(s) used	Fragment size(s)
A	*Pst*I	1399bp, 6182bp
B	*Pst*I	2832bp, 4749bp
C	*Eco*RI	2719bp, 4862bp
D	*Pst*I and *Eco*RI	643bp, 756bp, 2076bp, 4106bp

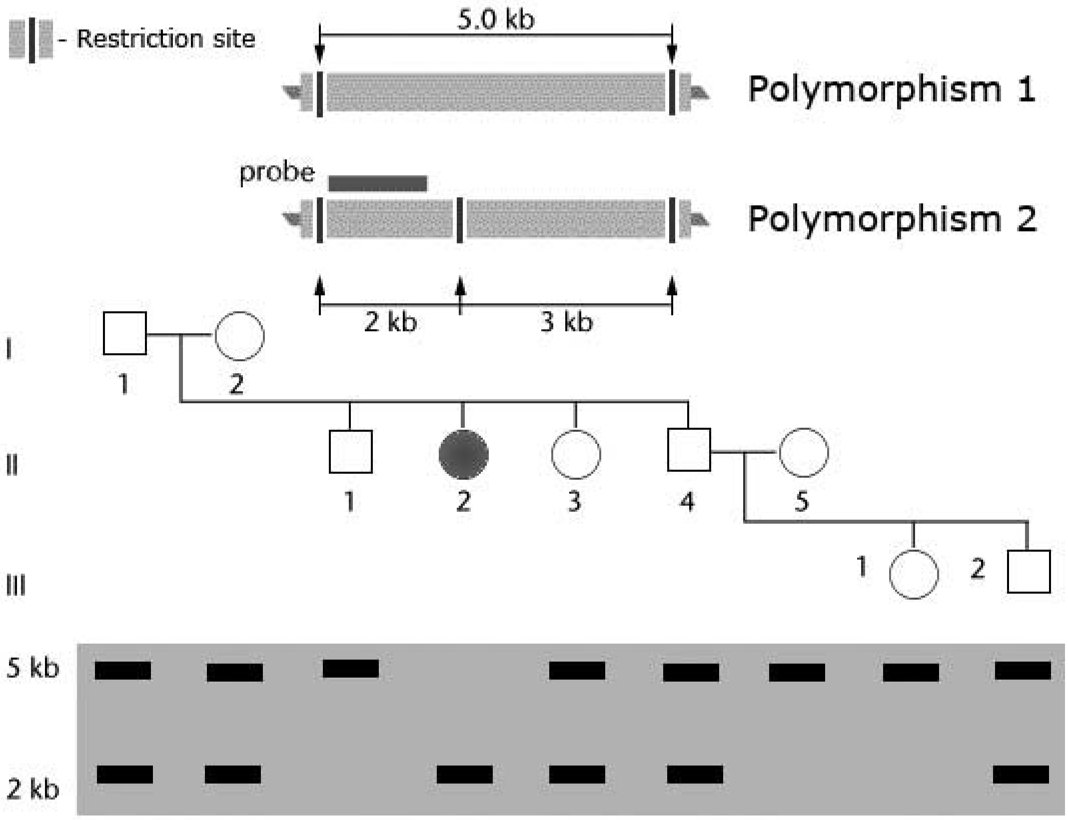
35 A sequence of DNA is to be amplified by the Polymerase Chain Reaction is shown below.

5' – ATTCTCGATCGGAAG [Target sequence] CGATCATATCGATAG – 3'
3' – TAAGAGCTAGCCTTC GCTAGTATAGCTATC – 5'

Which option shows suitable primers for amplifying this sequence?

	Primer 1	**Primer 2**
A	5' – ATTCTCGATCGG – 3'	5' – TCGATATGATCG – 3'
B	5' – CTTCCGATCGAG – 3'	5' – ATTCTCGATCGG – 3'
C	5' – GAAGGCTAGCTC – 3'	5' – GCTAGTATAGCT – 3'
D	5' – CCGATCGAGAAT – 3'	5' – TCGATATGATCG – 3'

36 Restriction Fragment Length Polymorphism (RFLP) was used to analyse the inheritance of a rare mutation that causes the pre-mature degeneration of the nervous system. A genetic marker on Chromosome 5 was discovered to be associated with the disease and used in the analysis. Genomic DNA was digested with a restriction enzyme *Bam*HI and the results of the gel electrophoresis and nucleic acid hybridisation are shown in the diagram below.



Which of the following observations cannot be made from the information provided?

A The mutation that causes the disease is probably recessive.

B The mutation that causes the disease is likely to be located in a coding region of the genome.

C The children of II-1 are unlikely to inherit the disease.

D There can only be a maximum of three *Bam*HI restriction sites on Chromosome 5 in the affected individual.

37 Cloning plants by tissue culture is more advantageous as compared to traditional vegetative propagation methods as

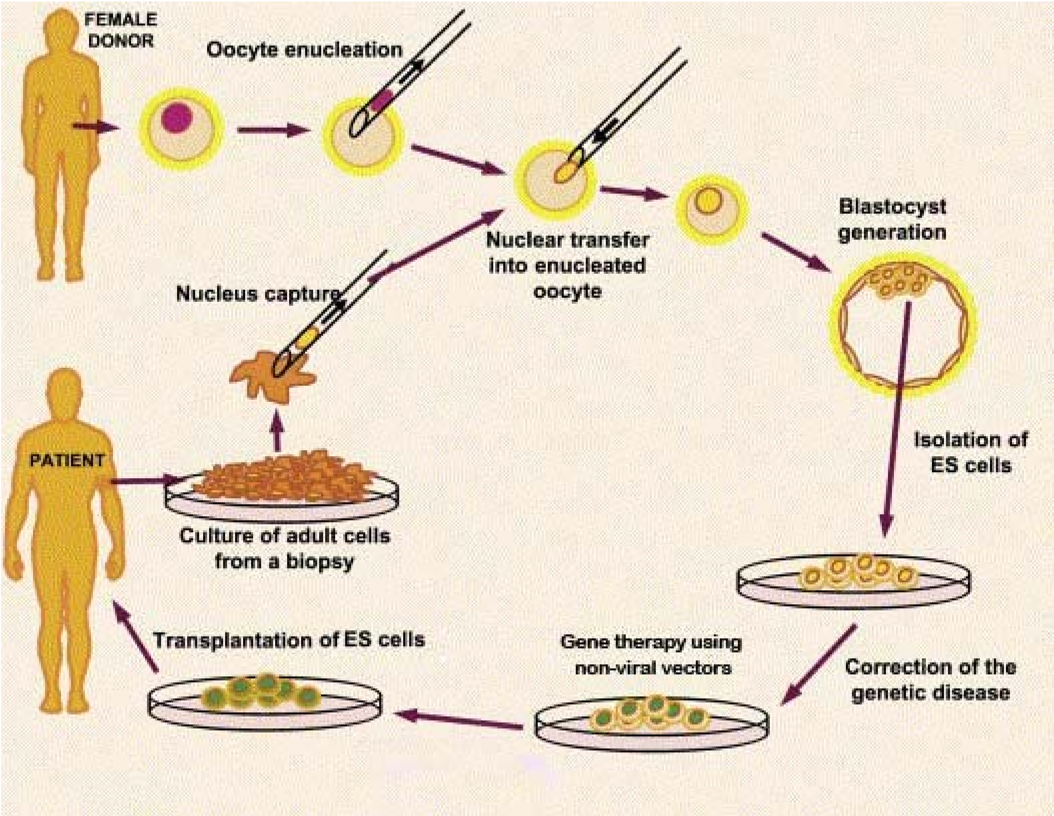
A The plantlets will require less space to grow to a saleable age.

B The plantlets can be grown regardless of seasonal variation.

C The plantlets will be genetically identical to the parent plant.

D The plantlets will be more resistant to fungi and diseases.

38 A proposed method of stem cell gene therapy is illustrated in the diagram below.



This treatment may not work because

A the patient's immune system recognises the transplanted cells as foreign particles and an immune response is triggered.

B only one copy of the normal allele is inserted to treat a recessively inherited disorder.

C the genetic disorder is caused by multiple genes.

D repeated transplants may be required.

39 Recent advances in the field of stem cell research have shown that induced pluripotent stem cells (iPS cells) can be artificially derived from adult somatic cells. iPS cells are mostly similar to natural pluripotent cells. This implies that

A iPS cells can theoretically differentiate into all cell types.

B iPS cells can theoretically differentiate into any of the three germ layers.

C iPS cells can theoretically differentiate into gametes.

D iPS cells are theoretically capable of transdifferentiation.

40 Which of the following theoretical application(s) of information obtained from the Human Genome Project is/are unethical?

I. Gene therapies for only certain genetic diseases are developed.

II. Potential parents undergoing genetic testing and genetic counselling prior to conceiving.

III. Potential parents choosing embryos for implantation only after ante-natal tests for selected genes.

IV. Increasing insurance premiums for people who are genetically predisposed to cancer.

V. A scientist studying phylogeny and early human migration patterns.

A I and V

B I and III

C II and III

D III and IV

THE END